



SUJET STATAPP 315 - 2026-2027

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Organisme : Ekimetrics

Catégorie : Traitement du signal / NLP

Sujet :

Efficient Evaluation of LLM-based Agents under a Limited Budget

Description du sujet :

LLM-based / AI agents are systems based on large language models (LLMs) that can interact with tools or an environment to complete multi-step tasks, such as modifying a software repository to solve a programming issue. Their behaviour is stochastic: running the same agent several times on the same task can lead to different outcomes. Evaluating an agent therefore often requires several independent attempts, called "rollouts", for each benchmark task.

These rollouts can be expensive, particularly for agents that make many model and tool calls. Current evaluation protocols commonly allocate the same number of rollouts to every task. Yet benchmark tasks can differ substantially in difficulty: some are almost always solved, while others succeed only rarely.

This raises a question of "statistical efficiency". An evaluation is more efficient if it can estimate the same performance metric with the same precision using fewer rollouts. In particular, an additional rollout on one task may provide much more information about the final benchmark score than an additional rollout on another task whose contribution is already well estimated.

The project asks:

> "Given a fixed number of agent rollouts, which benchmark tasks should be evaluated again in order to estimate agent performance as accurately as possible?"

The main performance measure will be "pass@k". If an agent has probability p of solving a task in one attempt,

p

$\text{pass@k} = 1 - (1-p)^k$

k

is the probability that it solves the task at least once in k independent attempts.

The benchmark contains a fixed collection of tasks. Students will first observe a small number of rollouts for each task, then investigate how the remaining evaluation budget should be distributed across these same tasks. At each stage, the question is therefore: "on which task would one additional rollout reduce uncertainty about the final pass@k estimate the most?"

Students will compare this adaptive strategy with uniform evaluation, where every task receives the same number of rollouts. They will study how the answer depends on task difficulty and on k , and whether the resulting performance estimates and confidence intervals remain reliable when previous outcomes influence where the next rollout is allocated.

A secondary analysis may consider "pass^k", the probability of succeeding on all k attempts, which measures consistency rather than best-of- k capability.

The students will "not build, train or run AI agents as part of the core project". They will analyse existing datasets containing many pre-generated repeated evaluations. The project is primarily about statistical estimation, experimental design and the efficient use of an evaluation budget.

Données du sujet :

Two public repeated-evaluation datasets will be used.

"ASSERT-KTH ? On Randomness in Agentic Evals:" 500 SWE-bench Verified software-engineering tasks, evaluated through 10 independent runs for 12 agent configurations (2 scaffolds $\times$ 3 models $\times$ 2 temperatures), giving approximately 60,000 trajectories with binary success/failure verdicts. Dataset: `ASSERT-KTH/agentic-evals-artifacts` (CC-BY-4.0).

"Large Language Monkeys:" up to 10,000 independent attempts per problem on GSM8K, MATH, CodeContests and MiniF2F for several language models. The large number of repetitions provides a high-precision reference estimate of each task's success probability and makes it possible to replay evaluation strategies under arbitrary smaller budgets. Dataset: `ScalingIntelligence/monkey\_business` (MIT).

Modèles envisagés du sujet :

NA. Methods will be compared using bias, variance, mean squared error, confidence-interval calibration and the number of rollouts required to reach a given estimation precision.

Bibliographie du sujet :

- Bjarnason, B. H., Silva, A., & Monperrus, M. (2026). On Randomness in Agentic Evals.
- Brown et al. (2024). Large Language Monkeys: Scaling Inference Compute with Repeated Sampling.
- Kossen, J., Farquhar, S., Gal, Y., & Rainforth, T. (2021). Active Testing: Sample-Efficient Model Evaluation. ICML.
- Jimenez, C. E. et al. (2024). SWE-bench: Can Language Models Resolve Real-World GitHub Issues? ICLR.
- Yao, S. et al. (2025). ?-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains ICLR.

Notation du sujet :

Commentaires :